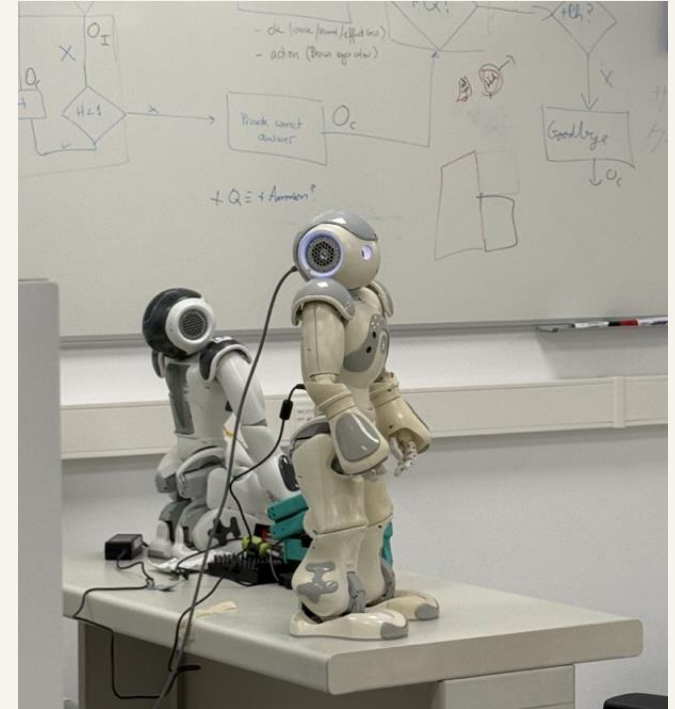


Design, Integration, and Validation of a Modular LLM-Based Interactive Architecture

A NAO-centred ROS 2 architecture for grounded dialogue, structured planning, deterministic execution, and evidence-based reporting.

Juan David Bendek Williamson • Supervised by Raquel Ros • 2025–2026





Language must become verifiable action

Language is flexible

- References depend on people, objects and prior turns.
- One instruction can combine dialogue, knowledge and action.
- A plausible answer may omit a target or invent a capability.

Execution needs explicit state

- Ground symbols in the available scene and knowledge.
- Validate capabilities, arguments and ordered steps.
- Establish completion from results, not from predicted effects.

Problem: connect flexible interpretation to bounded, inspectable execution.



Five questions guide the integrated stack.

How can a modular robot architecture integrate dialogue, symbolic grounding, planning, deterministic execution, and feedback while preserving clear responsibility boundaries and evidence-based communication?

RQ1–2 · Plan / ground

Can the stack preserve ordered tasks, target sets and the roles of people, objects and locations?

RQ3 · Admission

Can reviewed skills and deterministic checks reject unsupported or malformed proposals before dispatch?

RQ4–5 · Feedback

Can failures trigger recovery or truthful closure while responsibilities remain inspectable?

RQ1 plans

RQ2 grounding

RQ3 admission

RQ4 recovery

RQ5 ownership



Implementation and validation objectives

1

Integrate dialogue, routing, grounding, planning, orchestration and skills through ROS 2.

2

Define contracts for turn output, context, requests, skills, plans, results, feedback and evidence.

3

Expose capabilities through reviewed skill definitions, not generated robot API calls.

4

Support dialogue, knowledge and simple/composite actions with clear ownership.

5

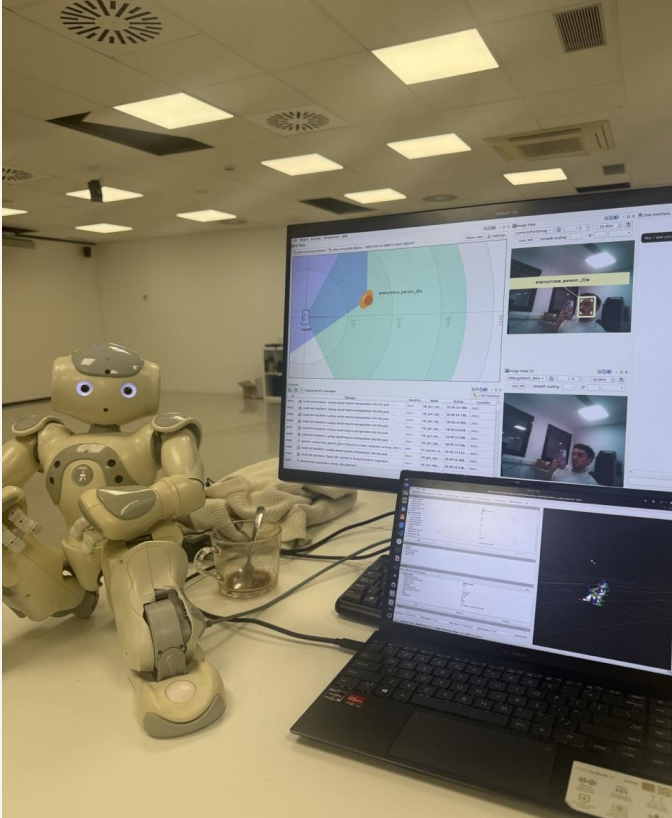
Validate candidate plans deterministically before dispatch.

6

Return typed evidence for failure, clarification, replanning and truthful reporting.

7. Validate full user turns, focused subsystems, controlled outcomes and symbolic postconditions.

NAO, ROS 2 and ROS4HRI



A humanoid connected through ROS nodes

NAO: speech, cameras, articulated motion and gaze.

ROS 2: independently running nodes with typed interfaces.

ROS4HRI: shared people, speech and interaction conventions.

TOPIC · publish/subscribe streams or events (.msg)
/scene/summary → std_msgs/msg/String

SERVICE · request → response (.srv)
DialogueInteraction → a dialogue turn

ACTION · goal → feedback → result; supports cancellation
/nao/say → communication_skills/action/Say

The ROS interface carries the message; the application contract defines its meaning.



An LLM agent is more than a model

Model + agent harness

LLM · large language model

A trained model generates text or structured output.

Multimodal models can also process images.

Agent harness

Supplies context, tools, state and output rules.

Feedback

Observations and results guide the next decision.

Two specialised agent roles

chatbot_llm

Interprets a turn, selects its route and prepares a grounded request.

planner_llm

Selects reviewed skills, proposes a plan and reacts to execution feedback.

Agent pattern: observe → propose → validate/act → inspect the result.



Grounded planning in embodied systems

Relevant approaches

PDDL / HTN

Explicit actions, conditions, effects and task decomposition.

SayCan · Ahn et al., 2022

Language-level skill selection linked to robot affordances.

PaLM-E · Driess et al., 2023

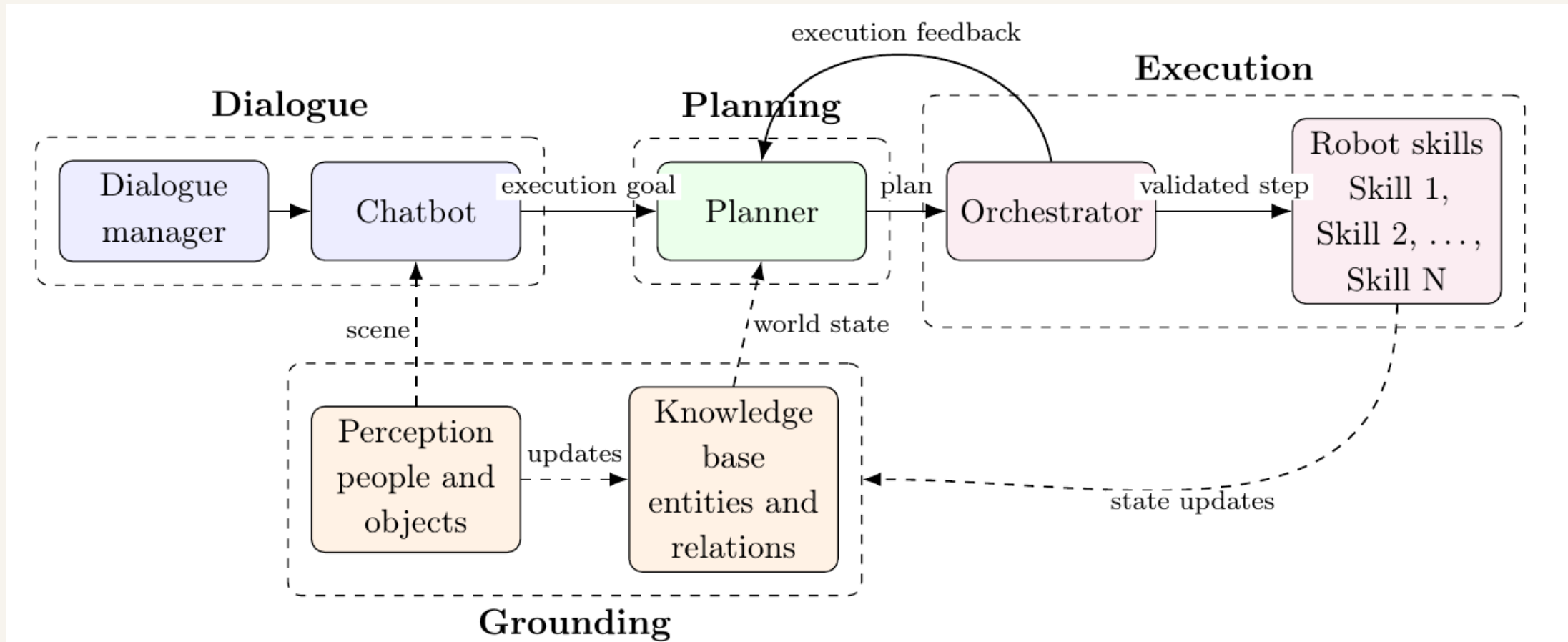
Multimodal inputs connected to embodied reasoning.

Position of this thesis

- Multimodal capability alone is not reliable grounding or execution.
- The stack uses compact symbolic context and reviewed skill contracts.
- Contribution: integrate language reasoning with validated execution and feedback—not train an end-to-end VLA.

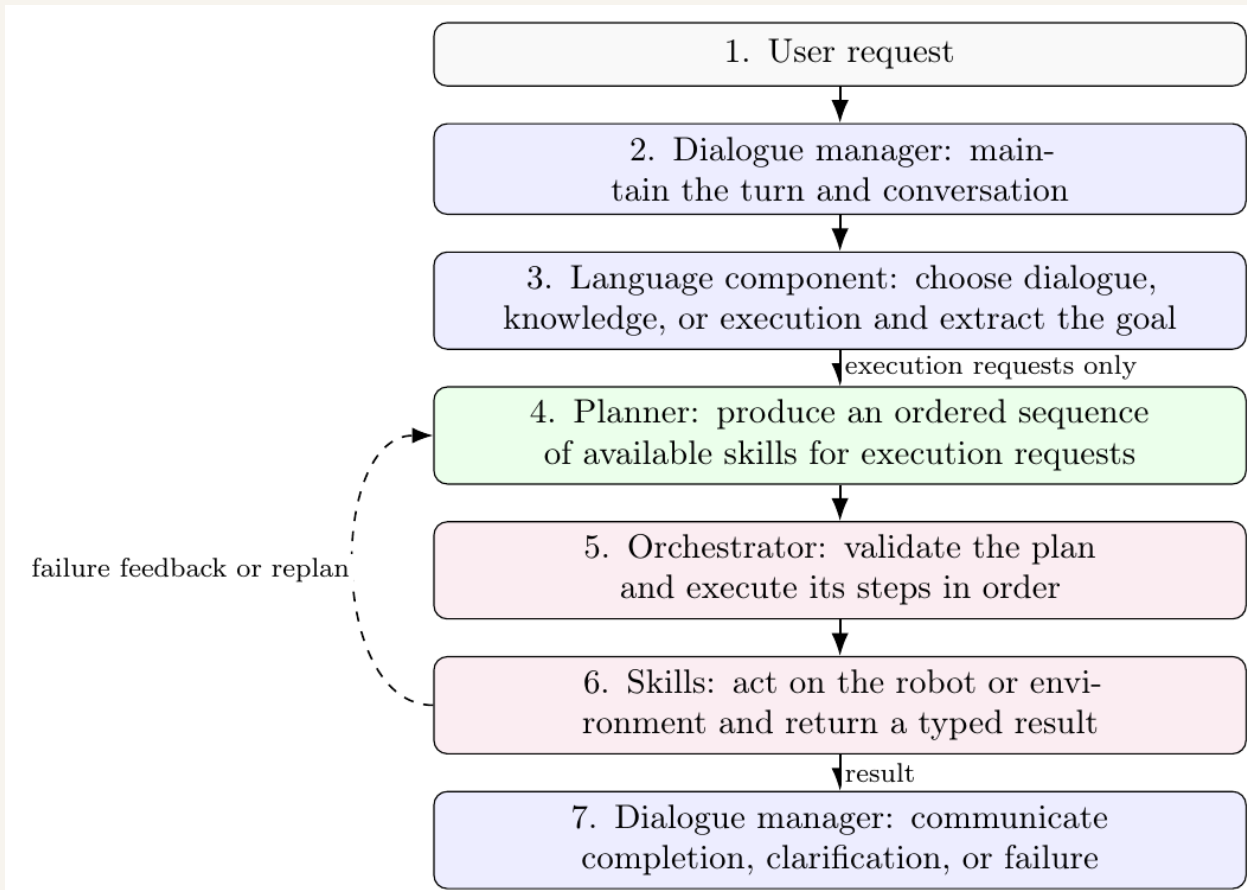
The planner interface connects flexible reasoning to explicit robot capabilities.

System architecture and ownership



Reused: dialogue, HRI and KnowledgeCore · Local: planning, orchestration and NAO integration

Runtime: request, action and feedback



Implementation adds explicit admission and dialogue-act relay boundaries.

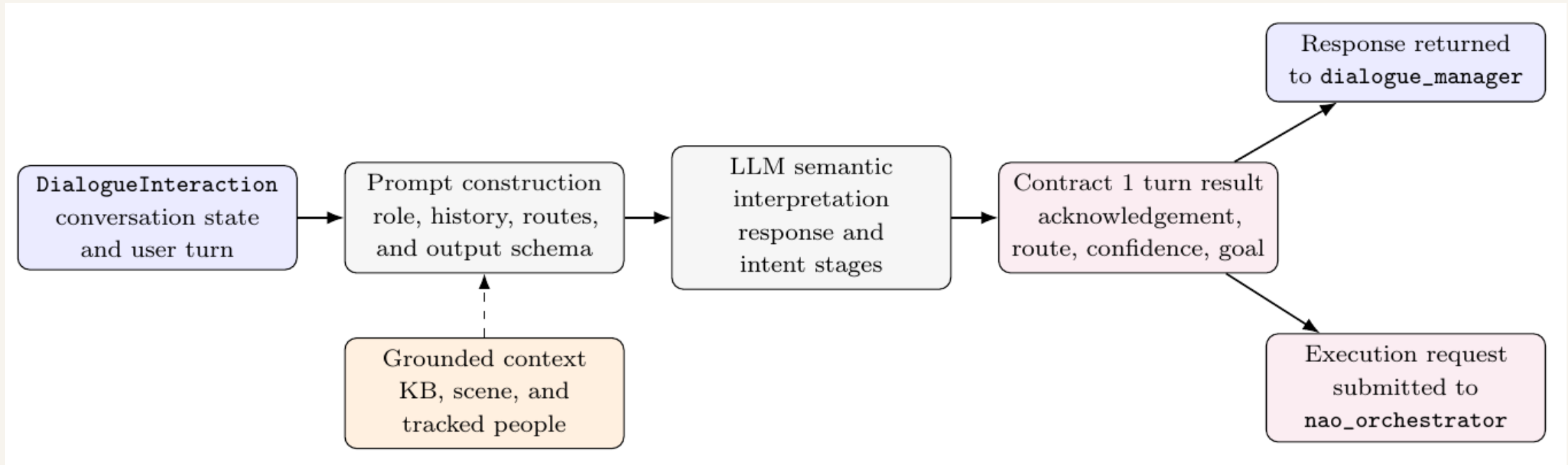
Execution-only planning

Dialogue / knowledge
Answer without a robot-action plan.

Execution
Admit the goal, generate a plan, validate and dispatch.

Feedback
Replan, clarify or close using observed results.

Dialogue ownership and chatbot routing



dialogue_manager · reused

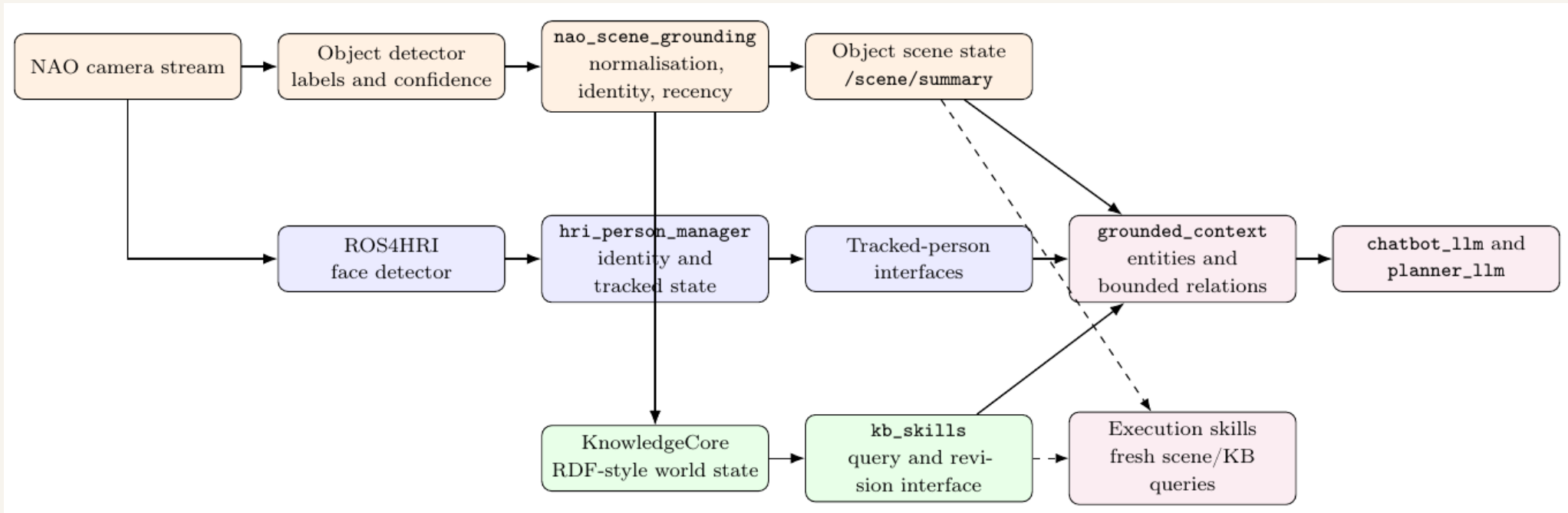
Owens conversation lifecycle and speaking. Calls `chatbot_msgs/srv/DialogueInteraction`.

chatbot_llm · extended

Returns `verbal_ack`, `route`, `confidence` and `user_intent`; submits execution requests, never executable plans.

`route` $\in$ {`dialogue`, `knowledge_query`, `execution`} · `goal_text` is an objective, not a step list.

Grounding builds a bounded scene context



Normalise and preserve roles

nao_scene_grounding: detector labels → stable IDs and transient facts. Reused HRI nodes retain person identity.

Project, do not duplicate the world

chatbot_llm merges scene/KB/person evidence into entities, locations and bounded relations. Raw evidence stays at its owner.

id = handle · label = readable name · kind/class = semantics · visible = current state · relations = context



The planner request is an admitted objective

Intent.data · abbreviated contract

```
{
  "dialogue_turn_id": "__default__:d343701c:1",
  "goal_id": "goal_default__d343701c_1",
  "request_kind": "new_goal",
  "goal_text": "look at ALEX and report
completion",
  "normalized_intents": ["look_at"],
  "grounded_context": {
    "entities": [
      {"id": "alex_person", "kind": "person",
"visible": true}
    ]
  }
}
```

Each field has a responsibility

dialogue_turn_id links to the conversation.
goal_id links plans and feedback.
request_kind proposes a goal transition.
goal_text states the objective.
normalized_intents guide skill selection.
grounded_context supplies bounded evidence.

Chatbot → orchestrator admission → planner.

/nao_orchestrator/planner_request → /planner/request
hri_actions_msgs/msg/Intent · JSON in data



A skill definition is the planner's vocabulary

pick_object · registry excerpt

```
{
  "object_id": "pick_object",
  "kind": "skill",
  "category": "manipulation",
  "aliases": ["pick", "grab", "grab_object"],
  "params": ["object_id", "target", "object",
    "support", "gripper", "result_mode"],
  "required_params": ["target"]
}
```

Identity is not an invocation

object_id: canonical capability name.
kind/category: skill and semantic family.
aliases: accepted compatibility names.
params: allowed argument names.
required_params: mandatory subset.

A plan invokes this capability with concrete args; the registry is its definition.

Registry object_id identifies the skill; args.object_id identifies a target entity.



Declared effects must become observed evidence

pick_object · semantics excerpt

```
{
  "preconditions": ["target object must be
grounded or explicitly scenario-owned", "robot
gripper should be free"],
  "expected_effects": ["robot holds the object",
"previous support relation is removed or
replaced"],
  "observable_success": ["status",
"evidence.held", "evidence.kb_effects"],
  "failure_modes": ["object_unavailable",
"unreachable", "already_held", "grasp_failed"]
}
```

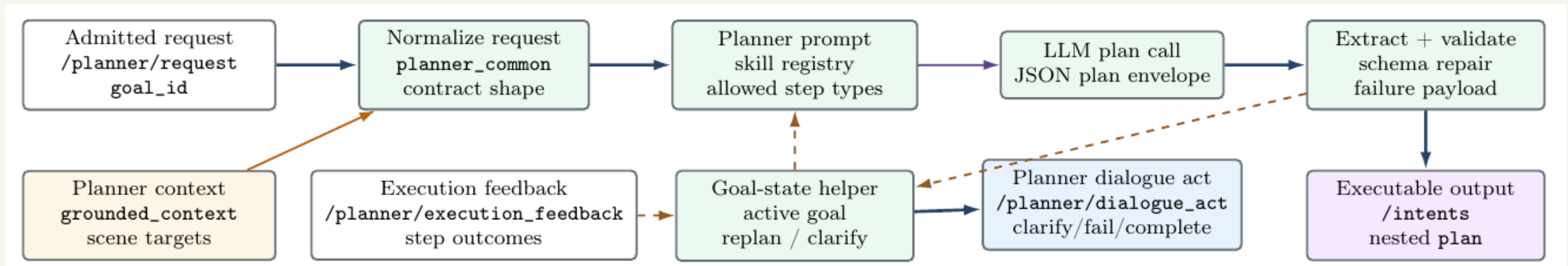
Selection, checking and dispatch

planner_guidance: selection and ordering.
result_schema: required result fields.
safety_flags: admission/dispatch labels.
runtime_callable: exposure gate.
implementation_status: real/fake/hybrid.
robot_adapter_mapping: execution adapter.
supports_failure_injection: test support.

preconditions: assumptions to check.
expected_effects: predicted state change.
observable_success: evidence to inspect.
failure_modes: known failure outcomes.

Definition → invocation → execution result: three different objects with different authority.

The planner proposes and supervises plans



A reviewed capability vocabulary

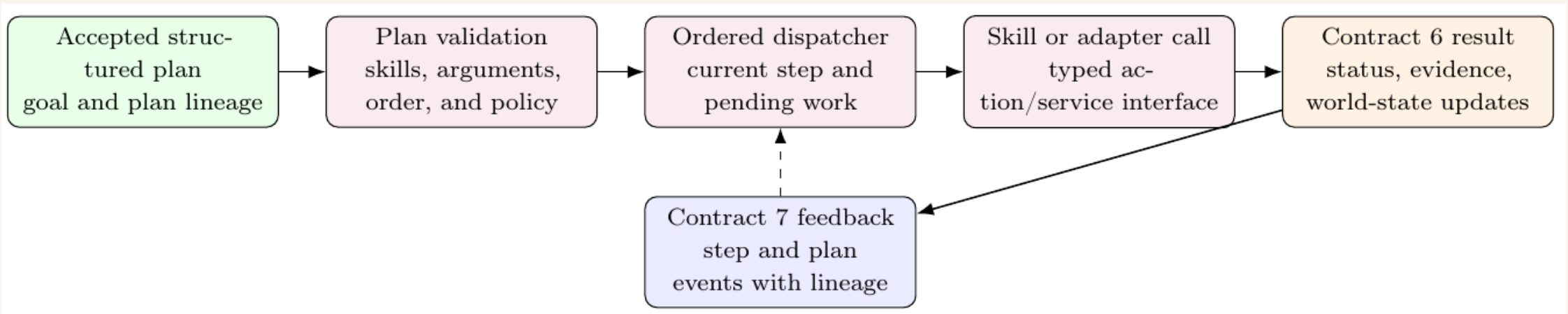
Admitted goal + grounded context + skill entries condition the model. Parse and validate the candidate before publication.

Lineage survives revision

plan.goal_id, plan_id and plan_version identify the proposal. steps contain id, name, args, requires and on_failure.

/planner/request → planner_llm → /intents · hri_actions_msgs/msg/Intent, JSON nested under plan

The orchestrator controls execution



Check before dispatch

Validate skill names, arguments, target roles, dependencies, policy and lineage. Track accepted and pending work.

Dispatch and verify

Call the configured skill interface. Correlate typed results; apply successful KB effects via `kb_skills` and check postconditions.

`nao_orchestrator` owns deterministic execution—not task-planning policy or natural-language generation.



Results drive feedback and truthful closure

Skill result · thesis excerpt

```
{
  "skill": "place_object",
  "status": "succeeded",
  "summary_text": "I placed cup_1 on table_1.",
  "evidence": {
    "object_id": "cup_1",
    "destination_id": "table_1",
    "kb_effects": [
      {"operation": "add", "statement": "cup_1
oro:isOn table_1"}
    ],
  },
  "failure": null,
  "metadata": {"fake": true}
}
```

Result → feedback → truthful closure

C6 · skill result
status + summary_text + evidence + failure + metadata.

C7 · executor feedback
goal/plan/version + event + step + status/reason + result_payload.

Reporting boundary
The planner supplies structured status.
Dialogue Manager alone owns speaking.

Action result → /planner/execution_feedback → planner supervision



Real and fake skills share a result boundary

Robot-facing adapters

Speech · /nao/say

Posture / head · motion adapters

Gaze · ROS4HRI-compatible look_at

Physical integration was exercised. The main semantic profile did not establish physical navigation, grasping or safety rates.

Controlled fake_skills

Action endpoints: /skill/fake/<skill>

Carrier: nao_skills/action/ScanScene

Scenario, success, failure, alternating or seeded policy; targeted fail-once profiles.

Real KB guards + simulated outcome + typed result + declared KB effects.

Keep metadata.fake=true visible; a simulated “held” result is not evidence of a physical grasp.



Evaluation qualifies a configuration

Controlled unit of analysis

One user request under a declared:

- model and generation settings;
- software, prompts and contracts;
- launch profile and selected adapters;
- symbolic scene and conversation state;
- harness and scoring procedure.

What the main profile isolates

Full-turn integration

Routing → planning → execution → closure.

Controlled skill outcomes

Repeatable success/failure for recovery.

Symbolic postconditions

State change checked independently from fluent speech.

Scenario-based software validation—not unrestricted autonomy or a human-subject study.



A pass requires the whole relevant trajectory.

A

Semantic route

Correct dialogue,
knowledge, or execution
path

B

Accepted plan

Complete targets, roles,
skills, order, and lineage

C

Controlled outcome

Success, failure,
clarification, or bounded
replan

D

Verified closure

Postcondition and final
report agree with
evidence

Route + plan + execution evidence + postconditions + truthful closure

Missing correlated evidence is not automatically a model failure; preserve exclusions explicitly.



Human-supervised, agent-assisted validation

Bounded development loop

robot-stack-iteration-loop

“Establish A Red Gate”

“Patch The Owing Seam”

“Accept a round only when source and live evidence agree.”

Delegated task → test/probe → proposed change
→ verification → human acceptance.

Evidence-based runtime review

robot-runtime-performance-review

“Read raw evidence, not summaries only.”

“Separate fact absence from model failure.”

“Score state and wording independently.”

Review routing, grounding, lineage, recovery and duplicate speech.

I thoroughly reviewed, revised and tuned all agent-generated project code.



One primary campaign; separate supporting sets.

108

Retained case records

33 source artifacts across the primary provenance index.

72

Primary records

Qwen3-VL: 64 passes and 8 non-passes under the primary qualification.

191

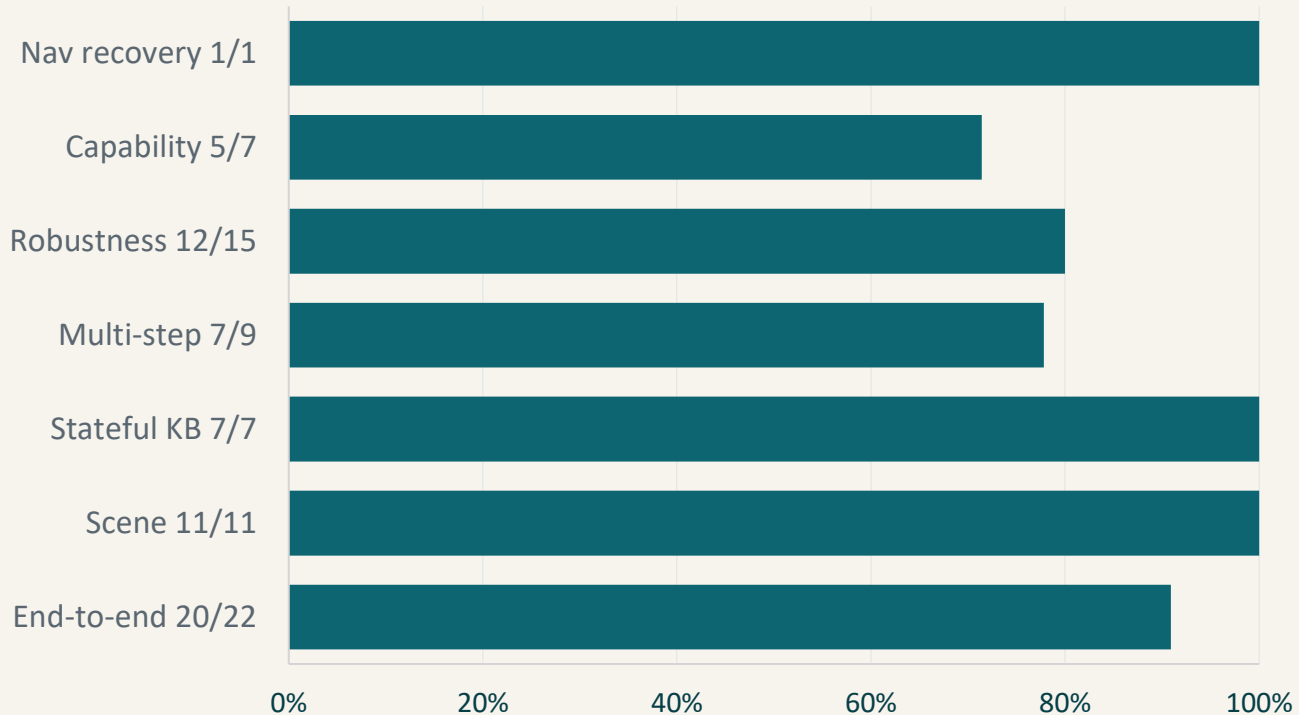
Model supplement

Three-model supplement. Kept outside the primary denominator.

108 = 72 primary + 26 ablation + 10 archived records

RESULTS

64 of 72 primary cases passed.



What the suites test

- Named scenes and relation grounding
- Stateful KB mutation and verification
- Five robustness prompts repeated three times
- Multi-step and multi-capability tasks

Interpretation

Controlled symbolic success is not physical performance. Safe rejection does not make an incomplete task successful.



Semantic failures remained attributable.

ROUTE	KB update entered the execution planner	Route and ownership failure
SELECTION	One member omitted from an “all objects” set	Target-set failure
ROLE	Kitchen treated as the delivery recipient	Entity-role failure
COVERAGE	Required action absent from a long composite	Plan-completeness failure
CLOSURE	Recovery occurred without reliable terminal speech	Report/closure failure

Harness and wiring defects were classified separately; fluent output was not sufficient evidence.



Shared interfaces; different semantic behaviour.

Stable interfaces

- Same scenario definitions
- Same ROS ownership
- Same symbolic transport
- Same request schema
- Same result contracts

Variable behaviour

- Route selection
- Target coverage
- Role assignment
- Clarification
- Plan and report completeness

Matched supplement

Comparable strict profiles:
Gemma4:31b-cloud 50/59
Gemma4:cloud 51/60
Nemotron-3-super 43/56

Descriptive results, not a statistical model ranking.

Backend compatibility permits substitution; each model still needs behavioural qualification.

A failed pick triggered a revised plan.



find → pick failed → feedback → scan / find / pick → report

Why this trace matters

The retained run completed after a middle-stage failure. The successful find was repeated. This is simulated recovery, not a physical grasp trial.



Three turns expose the main routes.

01

“Hey, how are you?”

one response, no planner request

Dialogue route

02

“What can you see now?”

answer from the declared symbolic state

Knowledge route

03

“Move your head up.”

accepted plan, one skill result, one closure

Execution route

Live demo illustrates the stack; frozen traces remain the scored evidence.



What the evidence supports

RQ1

Plans work within tested task families

Composite and multi-capability failures remain.

RQ2

Symbolic grounding supports role-aware tasks

Selection and role interpretation remain model-sensitive.

RQ3

Deterministic rejection contains tested errors

No general physical-safety or completeness guarantee.

RQ4

Feedback enables observed recovery

Beginning/middle evidence is stronger than final-stage recovery.

RQ5

Ownership and interfaces remain inspectable

Transfer to other robots still requires validation.



Strong software evidence; limited physical claims.

Supported

- ownership and interface architecture
- deterministic admission and dispatch
- controlled failure and replanning
- symbolic postconditions and reporting

Not established

- navigation or grasp accuracy
- detector performance
- human safety or acceptance
- unrestricted autonomy or universal model ranking

Transfer conditions

Another robot needs an equivalent reviewed skill registry and typed adapters. Another model must repeat the frozen qualification.

Controlled outcomes isolate software behaviour; they do not replace physical trials.



Future work: strengthen capability and evidence.

Semantic controls

- Fine-tuning / LoRA holdouts
- Cardinality, role and coverage checks
- KB-only routing by construction
- Require terminal closure

Embodied validation

- Measure perception and skills
- Fresh execution-time evidence
- Physical failure / cancellation
- Validate a second platform

Reproducible evidence

- Immutable source/run artifacts
- Independent repeated campaigns
- Separate model, software, harness and physical outcomes

Promote capability only with evidence appropriate to the claim.



Architecture, execution contracts and validation.

01

Architecture

Integrated dialogue, grounding, planning, orchestration and robot adapters with explicit ownership.

02

Containment

Reviewed skill definitions, typed payloads and deterministic checks constrain model-generated proposals.

03

Qualification

Controlled cases, retained traces, symbolic postconditions and human-supervised agent-assisted iteration.

An inspectable integration—not a new foundation model or a proof of unrestricted autonomy.



Language-model dialogue and planning can participate in an integrated robot architecture when their outputs are grounded, validated against reviewed skills, dispatched deterministically, and checked against typed feedback and symbolic postconditions.

The boundaries do not eliminate non-determinism. They make it observable and restrict its execution authority.

Thank you. I welcome your questions.



Short answers for likely questions

Why simulated skills?

They place repeatable outcomes at the robot-facing result boundary, isolating planning and recovery from hardware variance.

What is deterministic?

Admission, schema checks, dispatch order, lineage, feedback correlation, and effect verification. Route and plan generation remain model-dependent.

Why model-agnostic?

The interfaces accept multiple backends. The results explicitly show that semantic behaviour is not invariant.

Why no classical planner baseline?

The thesis validates integration and containment, not a causal comparison of planning algorithms.

What comes next?

Quantified physical skills, detector-quality evaluation, and user-facing interaction trials under the same evidence contracts.

Detailed primary results

Suite	Pass	Non-pass	Total / pass rate
End-to-end	20	2	22 cases · 90.9%
Scene / grounding	11	0	11 cases · 100%
Stateful KB	7	0	7 cases · 100%
Multi-step	7	2	9 cases · 77.8%
Robustness	12	3	15 cases · 80.0%
Multi-capability	5	2	7 cases · 71.4%
Navigation recovery	1	0	1 case · 100%

Main questionnaire · 1/5

Dialogue	Hey, how are you? One natural response; no planner request.
----------	--

Dialogue	What is your favourite movie? Answer without planner handoff.
----------	--

Dialogue	My favourite movie is Back to the Future! Use conversation context; no execution.
----------	--

Dialogue	Do you have any ideas for this weekend? Advice only.
----------	---

Dialogue	What is the speed of light? Informational answer only.
----------	---

Main questionnaire · 2/5



Knowledge

What can you see now?
Use declared current state.

Knowledge

What new objects can you see?
Use newly inserted scene facts.

Knowledge

Which object is on the table?
Resolve the support relation.

Knowledge

What is the cup called?
Return the stored name.

Knowledge

How many objects can you see?
Count deliverable objects, not people/support anchors.

Main questionnaire · 3/5



Knowledge	What colour is the cup? Return the stored colour.
Knowledge	Add a red cup to your knowledge base. Authorised KB mutation, no robot-action plan.
Execution	Move your head up. Execute motion and report result.
Execution	Pick up the phone. Ground one phone before pick.
Execution	Pick up the object on the table. Select from support relation.

Main questionnaire · 4/5



Execution

Look at ALEX.
Preserve human gaze-target role.

Execution

Move your head up, down, left, and right.
Preserve all motions and order.

Execution

Bring all the objects you can see to ALEX.
Preserve target set and human recipient.

Execution

Go to the kitchen, find the cup, and bring it to me.
Preserve location, object, recipient and order.

Execution

Go to ALEX, wave, and tell me when you are finished.
Navigate, perform greeting motion, report.

Main questionnaire · 5/5

Execution

Visit every object and tell me when you reach each one.
Visit each object and report each arrival.

Dialogue

Remember that ALEX is the recipient, but do not act yet.
Memory acknowledgement without action.

Scene and grounding · 1/3

**baseline table
inventory**

What objects are on the table?

lab table inventory

What objects are on the table?

lab sections inventory

What can you see in the lab?

lab sections delivery

Bring every object from the work table to ALEX and report what happened.

Scene and grounding · 2/3

kitchen inventory

What is in the kitchen?

**grouped location
delivery**

Bring every object from the kitchen to ALEX and report what happened.

**grouped location
followup**

Where is the kitchen cup now?

iiia floor inventory

What rooms and objects are in the IIIA floor scene?

Scene and grounding · 3/3

iiia kitchen delivery Bring every object from the kitchen to ALEX and report what happened.

gold apple handoff Can you bring that gold apple to the person named ALEX?

gold apple followup Where is the gold apple now?

Stateful KB · 1/2

seed inventory What are the names, colors, and current locations of the two objects in the test area?

revise support The cup has been moved. Which surface is TITAS on now, and where is MIDAS?

revised relation query Which object is on the storage shelf, and which object remains on the work table?

grounded delivery Bring TITAS from the storage shelf to the person named ALEX and report what happened.

Stateful KB · 2/2

delivery postcondition

Where are TITAS and MIDAS now?

move remaining object

MIDAS is now on the storage shelf. Bring every object on that shelf to ALEX and summarize the overall result.

mixed final query

Summarize the names, colors, and current locations of TITAS and MIDAS after those changes.

Controlled multi-step · 1/3

baseline inventory	What objects are on the table?
ordered walk report	Walk to every object on the table and let me know when you get to each one.
grouped work table delivery	Bring every object from the work table to ALEX and report what happened.
missing object recovery	Find the codex missing cup. If you cannot find it, scan the scene and report what you can confirm.

Controlled multi-step · 2/3

missing recipient clarification	Bring every object from the work table to the person named BLAKE and report what happened.
iiia kitchen delivery	Bring every object from the kitchen to ALEX and report what happened.
iiia floor location followup	Where are the objects from the kitchen now?
gold apple multiturn	Can you bring that gold apple to the person named ALEX?

Controlled multi-step · 3/3

gold apple followup

Where is the gold apple now?

Robustness · 1/2

paraphrased group delivery	Take everything resting on the work table over to ALEX. Once every item has reached them, summarize the whole delivery.
object role exclusion	Visit each movable item on the table in turn. Do not navigate to the table or to any person, and report each arrival.
missing recipient	Deliver every work-table object to MORGAN and explain what information is missing if MORGAN is not present.
future recipient memory	ALEX will receive the work-table items later. For now, only acknowledge this and do not execute anything.

Robustness · 2/2

**multiturn execute
followup**

Now deliver all of those objects to them and tell me the overall result.

Multi-capability · 1/2

walk pick sit report Walk to TITAS, pick it up, sit down, and summarize the result.

kneel under table pick report Kneel down, pick up MIDAS from under the work table, stand, and summarize the result.

dialogue inventory Before we do anything, tell me which named objects and people are in the test area.

dialogue sit stand grab return Now sit, stand again, pick up MIDAS, return to ALEX, and summarize the result.

Multi-capability · 2/2

all objects visit look wave sit	Stand up, walk to each visible object on the work table, look at each one, wave to ALEX, sit down, and give one final summary.
pick place kneel report	Pick up ATLAS, place it on the storage shelf, kneel, and summarize the result.
unreachable object recovery	Kneel and try to pick up the heavy box under the table. If you cannot, recover safely and summarize the result.



Contract 1: chatbot turn output

Explanatory shape—not a recorded event

```
{
  "verbal_ack": "I will look at ALEX.",
  "route": "execution",
  "confidence": 0.82,
  "user_intent": {
    "type": "look_at",
    "goal_text": "look at ALEX",
    "scene_targets": ["ALEX"],
    "request_kind": "new_goal"
  }
}
```

Field interpretation

verbal_ack: immediate response wording.
route: dialogue / knowledge_query / execution.
confidence: model/stage estimate, not proof.
user_intent.type: normalised label.
goal_text: user objective.
scene_targets: hints, not validated IDs.
request_kind: proposed goal transition.

No executable skill sequence is hidden inside chatbot intent.



Contract 2: grounded context

Thesis entity excerpt

```
{
  "id": "cup_jrjic",
  "label": "cup",
  "kind": "object",
  "class": "Cup",
  "visible": true,
  "relations": [
    {"predicate": "dbp:color", "object": "blue"},
    {"predicate": "oro:isOn", "object":
"table_1"},
    {"predicate": "oro:isAt", "object":
"lab_area"}
  ]
}
```

Bounded representation

id: stable target handle.
label: readable description.
kind: object / person / location.
class: semantic category.
visible: currently accepted scene/KB state.
relations: task-relevant predicates.

locations is a derived grouping view; people never become deliverable objects.

A current context supports planning; the executing skill must obtain fresh evidence.



Skill metadata and dispatch

Exact selected registry fields

```
{
  "planner_guidance": ["use for explicit pick or
grab requests after resolving a grounded object
id"],
  "result_schema": {
    "required": ["skill", "status",
"summary_text", "evidence"]
  },
  "safety_flags": ["manipulation",
"object_contact"]
}
```

How the fields are used

Planner guidance constrains selection.
Result schema defines evidence obligations.
Safety flags label relevant hazards.
Runtime callability gates exposure.
Implementation status names the backend class.
Adapter mapping resolves the execution owner.
Failure-injection support enables controlled tests.

The local canonical-format mirror agrees with the planner projection for pick_object.



Skill implementation and callability

Exact selected registry fields

```
{
  "object_id": "pick_object",
  "kind": "skill",
  "implementation_status": "fake",
  "robot_adapter_mapping":
  "fake_skills.pick_object",
  "supports_failure_injection": true,
  "metadata": {"runtime_callable": true}
}
```

How the fields are used

Planner guidance constrains selection.
Result schema defines evidence obligations.
Safety flags label relevant hazards.
Runtime callability gates exposure.
Implementation status names the backend class.
Adapter mapping resolves the execution owner.
Failure-injection support enables controlled tests.

The local canonical-format mirror agrees with the planner projection for pick_object.



Contract 5: executable plan

Thesis plan excerpt

```
{ "plan": {  
  "goal_id": "goal_turn_123",  
  "plan_id": "plan_123", "plan_version": 1,  
  "status": "planned",  
  "steps": [{  
    "id": "step_1", "type": "skill",  
    "name": "look_at",  
    "args": {  
      "target_frame": "anonymous_person_ehfbf"  
    },  
    "requires": [], "on_failure": "replan"  
  }]  
}}
```

Interpretation

goal_id: objective lineage.
plan_id: concrete proposal identity.
plan_version: revision.
status: planning outcome, not execution success.
steps: ordered invocations.
id: step correlation key.
args: schema-valid arguments.
requires: successful dependencies.
on_failure: validated recovery policy.

All plan metadata stays nested under plan; retry budgets remain separate from failure policy.



Contracts 7–8: feedback and communication

Execution feedback · thesis excerpt

```
{
  "goal_id": "goal_turn_123",
  "status": "failed",
  "step": {"id": "step_1", "name": "look_at"},
  "reason": "target frame unavailable",
  "result_payload": {"target_found": false,
  "target_kind": "person"}
}
```

Dialogue act · thesis excerpt

```
{
  "goal_id": "goal_turn_123",
  "act": "ask_clarification",
  "await_user_response": true,
  "reason": "multiple visible people",
  "text_hint": "Which person should I look at?",
  "slots_needed": ["target_person"]
}
```

Two separate illustrative excerpts—not one correlated runtime event.



ROS action carrier and fake configuration

ScanScene.action · actual definition

```
string target
string target_kind
int32 max_sweeps
string evidence_policy
string result_mode
---
bool success
string message
string summary_text
string result_payload_json
float32 duration
---
string status
float32 progress
```

Selected configuration fields

```
perform_motion_execution_mode:=real
look_at_execution_mode:=fake
start_fake_skills:=true
```

```
global_mode: scenario / always_success /
always_fail / every_other / random_seeded
active_scenario_id: named scenario
mode_overrides_json: per-skill policy
deterministic_seed: repeatable sampling
```

Result JSON is carried in result_payload_json.

The selected adapter and policy are part of the experimental configuration.



KnowledgeCore: state and verified effects

RDF-style knowledge statements

```
codex_lab_cup rdf:type Cup
codex_lab_cup dbp:color red
codex_lab_cup oro:isOn codex_lab_table
person_alex rdf:type Human
myself sees person_alex

read: /kb/query
mutate: /kb/revise
```

Ownership and verification

KnowledgeCore is reused infrastructure.
kb_skills owns query/revision transport.

Perception supplies transient observations.
Fixtures declare controlled symbolic state.
Skills report their observed effects.
The orchestrator applies successful effects and
queries required postconditions.

A correct KB postcondition and correct user-facing wording are separate checks.



Recovery trace: retained event sequence

Correlated outcome events

```
1783979352.183  plan_1783979351305
step_succeeded find_object
1783979352.946  plan_1783979351305
step_failed pick_object
1783979359.641  plan_1783979355769
step_succeeded scan
1783979360.506  plan_1783979355769
step_succeeded find_object
1783979361.899  plan_1783979355769
step_succeeded pick_object
1783979364.016  plan_1783979355769
step_succeeded report_result
1783979364.017  plan_1783979355769
plan_completed
```

Read the evidence carefully

Request: “Pick up the phone.”
Goal: goal_default___718b0d50_2

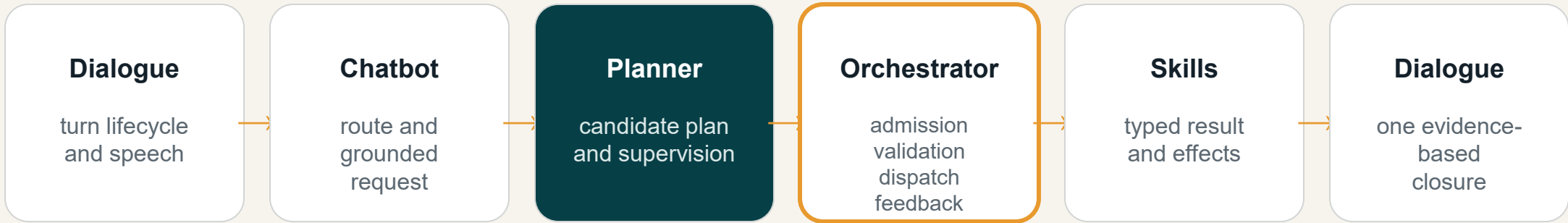
Initial plan ends with failed pick.
Replacement scans, finds, picks and reports.
The successful find is repeated.

Historical F13 case; not an extra row in the 72-case
primary campaign.

Derived display of retained log fields; raw artifact and exact source lines are in speaker notes.



Owners and evidence remain separate.



Goal ID • plan ID • plan version • stable step IDs

Symbolic evidence

KnowledgeCore and scene grounding provide bounded entities, roles, locations, and relations.

Execution evidence

Robot-facing or simulated adapters return the same typed result boundary.



Evidence and interpretation notes

Primary vs supplementary evidence

108 primary-index records:
72 qualification + 26 ablations + 10 archive.

72 qualification: 64 pass, 8 non-pass.

191 separate model-supplement records:
150 pass, 32 fail, 9 excluded.

Repeated prompts are not independent user trials.

Boundaries of reproducibility

Retain model/configuration, case identity, world fixture, source hashes and raw evidence.

A successful diagnostic rerun does not erase a retained non-pass.

Missing terminal evidence can fail acceptance; external/harness absence may be excluded under the declared scoring rule.

Sources: final thesis Tables 6.1–6.8 and Appendix D; retained JSON/CSV artifacts.